

Assignment 6

Given Data.

Machine	Temperature	Vibration	Failure.
1	High	High	Yes
2	High	Low	Yes
3	High	High	No
4	Medium	Low	No
5	Low	Medium	No
6	Low	Low	No
7	Medium	Medium	Yes
8	High	High	No

$$\{ \text{Entropy} = \sum p_i \log_2 p_i \}$$

No. of yes = 4

$$p(\text{yes}) = 4/8$$

No. of no = 4

$$p(\text{No}) = 4/8$$

$$\begin{aligned} \text{Entropy} &= -4/8 \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8} \\ &= \frac{4}{8} + \frac{4}{8} = 1 \end{aligned}$$

* When Temperature is High:

$$p(\text{yes}) = 3/3 = 1$$

$$p(\text{No}) = 0/3 = 0$$

$$\text{Entropy} = -(1) \log_2 (1) - 0 \log_2 0 = \underline{0}$$

* When Temperature is Medium:

$$p(\text{yes}) = 1/2$$

$$p(\text{No}) = 1/2$$

$$\text{Entropy} = -1/2 \log_2 (1/2) - 1/2 \log_2 1/2$$

$$= -1/2 \times 1 - 1/2 \times 1$$

$$= 0.5 + 0.5 = 1$$

* When the temperature is low:

$$p(\text{yes}) = 0$$

$$p(\text{no}) = 3$$

$$\begin{aligned}\text{Entropy} &= -(0) \log_2(0) - (1) \log_2(1) \\ &= 0 - 1(0) \\ &= \underline{\underline{0}}\end{aligned}$$

$$\begin{aligned}\text{Weighted Entropy} &= \frac{3}{8} \times 0 + \frac{2}{8} \times 1 + \frac{3}{8} \times 0 \\ &= 0 + 0.25 + 0 = \underline{\underline{0.25}}\end{aligned}$$

$$\begin{aligned}\text{* Information gain of temperature} &= \text{Entropy} - \text{Weighted Entropy} \\ &= 1 - 0.25 \\ &= \underline{\underline{0.75}}\end{aligned}$$

* For Vibration;

→ When vibration = High

$$p(\text{yes}) = \frac{2}{3}$$

$$p(\text{no}) = \frac{1}{3}$$

$$\begin{aligned}\text{Entropy} &= -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} \\ &= \underline{\underline{0.918}}\end{aligned}$$

⇒ When vibration = medium.

$$p(\text{yes}) = \frac{1}{2}$$

$$p(\text{no}) = \frac{1}{2}$$

$$\begin{aligned}\text{Entropy} &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ &= -\frac{1}{2}(-1) - \frac{1}{2}(-1) \\ &= 0.5 + 0.5 = \underline{\underline{1}}\end{aligned}$$

⇒ When vibration is low;

$$p(\text{yes}) = \frac{1}{3}$$

$$p(\text{no}) = \frac{2}{3}$$

$$\begin{aligned}\text{Entropy} &= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \left(\frac{2}{3}\right) \\ &= 0.528 + 0.390 \\ &= \underline{\underline{0.918}}\end{aligned}$$

Weighted Entropy of Vibration =

$$= \frac{3}{8} (0.918) + \frac{3}{8} (0.918) + \frac{2}{8} (1)$$

$$= \underline{0.9385}$$

$$IG(\text{Vibration}) = 1 - 0.9385$$

$$= \underline{0.0615}$$

{ IG of temperature > IG of Vibration }

Root node = Temperature.

ie, Decision Tree

